

Incident Report Analysis

This is a Security Incident Report for a fictional scenario given as an activity for the *Connect and Protect: Networks and Network Security Course* in Coursera.

Scenario:

You are a cybersecurity analyst working for a multimedia company that offers web design services, graphic design, and social media marketing solutions to small businesses. Your organization recently experienced a DoS attack, which compromised the internal network for two hours until it was resolved.

During the attack, your organization's network services suddenly stopped responding due to an incoming flood of ICMP packets. Normal internal network traffic could not access any network resources. The incident management team responded by blocking incoming ICMP packets, stopping all non-critical network services offline, and restoring critical network services.

The company's cybersecurity team then investigated the security event. They found that a malicious actor had sent a flood of ICMP pings into the company's network through an unconfigured firewall. This vulnerability allowed the malicious attacker to overwhelm the company's network through a denial of service (DoS) attack.

To address this security event, the network security team implemented:

- A new firewall rule to limit the rate of incoming ICMP packets
- Source IP address verification on the firewall to check for spoofed IP addresses on incoming ICMP packets
- Network monitoring software to detect abnormal traffic patterns
- An IDS/IPS system to filter out some ICMP traffic based on suspicious characteristics

As a cybersecurity analyst, you are tasked with using this security event to create a plan to improve your company's network security, following the National Institute of Standards and Technology (NIST) Cybersecurity Framework (CSF). You will use the CSF

to help you navigate through the different steps of analyzing this cybersecurity event and integrate your analysis into a general security strategy.

Summary	The multimedia company experienced a denial of service (DoS) attack that stopped its internal network from working for two hours. A malicious actor sent a flood of ICMP packets (pings) into the network through a firewall that had not been configured. The flood overwhelmed the network, so normal internal traffic could not reach network resources. The incident management team blocked incoming ICMP packets, took all non-critical network services offline, and restored critical network services. The cybersecurity team later found that the unconfigured firewall was the vulnerability the attacker used.
Identify	**Attack type:** ICMP flood, a type of denial of service (DoS) attack, in which a malicious actor overwhelmed the network with ICMP ping packets. **Source:** An external malicious actor. The scenario does not state the source IP address. **Vulnerability:** A firewall that was not configured to filter or limit incoming ICMP packets. **Affected systems:** The company's internal network, its network services and resources, and the firewall. **Impact:** Normal internal traffic could not access network resources for two hours.
Protect	The team added a new firewall rule to limit the rate of incoming ICMP packets, source IP address verification on the firewall to check for spoofed IP addresses on incoming ICMP packets, and an IDS/IPS system to filter out some ICMP traffic based on suspicious characteristics. To further protect the network, the team should review and update all firewall rules on a regular schedule, keep firewall and network device software up to date, and confirm that any new network device is configured before it is connected.
Detect	The team uses network monitoring software to detect abnormal traffic patterns, such as a sudden rise in ICMP packets. The IDS/IPS system flags ICMP traffic with suspicious characteristics, and source IP address verification helps

	identify spoofed addresses. The team should review firewall logs and alerts for incoming ICMP packets from untrusted external IP addresses, set an alert for when ICMP traffic goes above a set level, and review user account activity to spot unauthorized access.
Respond	**During this incident:** The incident management team contained the attack by blocking incoming ICMP packets and taking all non-critical network services offline so the network could recover. Critical network services were restored first. The cybersecurity team then investigated and found that the unconfigured firewall had been exploited. **For future incidents:** Contain the attack by blocking the malicious traffic at the firewall and isolating affected devices or network segments. Neutralize it by applying the firewall rules and IDS/IPS filtering. Collect data for analysis, such as firewall logs, network traffic logs, and IDS/IPS alerts. Report the incident to management, then hold a review afterward to update the response plan.
Recover	Restore critical network services first, then bring non-critical services back online step by step once the flood has stopped and the new firewall rules are confirmed to be working. To recover quickly, the team needs a list of critical and non-critical services with the order to restore them, and saved copies (backups) of the firewall and network device configurations. After recovery, test that services work normally and monitor network traffic to confirm the attack has not resumed. Finally, document the incident and update the response plan.

Reflections/Notes: